MMM-F-1486B
November 9, 1978
SUPERSEDING
Fed. Spec. MMM-F-1486A
May 18, 1976

FEDERAL SPECIFICATION

FILLER, PIPET, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of rubber pipet filler suitable for use with pipets in the transfer of corrosive, toxic, infectious and odorous liquids.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Bulb and pipet receptacle. The bulb and pipet receptacle of the filler shall be made of a rubber compound containing a minimum of 50 percent elastomer. The elastomer shall comprise a minimum of 99 percent pure natural rubber (derived from the gum of the rubber tree).

3.1.2 Ball valve. The ball valve shall be made of polished pure mineral agate, or polished corrosion resistant steel, AISI 300 series.

3.2 Design and construction. The pipet filler shall be in accordance with figure 1, except that alternate section A-A shall apply in lieu of section A-A. The item shall consist of a hollow spherical bulb, three ball valves and a pipet receptacle. The throat of the pipet filler shall fit snugly on pipets with upper end diameters of 0.092 to 0.30 inches. The valves shall operate independently and shall be capable of accurately controlling the inflow of liquids into the pipet, holding the desired level, and of accurately controlling the discharge.

3.3 Performance.

3.3.1 Ball valve. The ball valves shall exhibit no leakage, after stabilization (initial surge), when tested as specified in 4.4.1.

3.3.2 Aging. The rubber portion of the filler shall not stiffen, soften, crack, or become tacky when tested as specified in 4.4.2.

3.4 Color. Unless otherwise specified (see 6.2), the pipet filler shall be black.

3.5 Marking.

3.5.1 Identification. The pipet filler shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.5.2 Valve. The letters A, S and E shall be legibly molded into both sides of the valves as specified in figure 1.

3.6 Instructions. Printed instructions for use of the filler shall accompany each unit.

3.7 Workmanship. Workmanship shall be first class throughout. The pipet filler shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for examination purposes shall be one pipet filler.

TABLE I. Sampling for examination

Type of defect	Inspection level	AQL (defects per hundred units)
For visual examination		
Major A defects	S-2	1.0
Major B defects	S-2	1.0
Minor defects	S-2	2.5
For dimensional examination	S-1	1.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for test purposes shall be one pipet filler. Acceptance number shall be zero for all sample sizes.

TABLE II. Sampling for tests

Characteristic	Requirement	Test procedures	Inspection level	AQL (defects per hundred units)
Ball valves	3.3.1	4.4.1	S-1	1.0
Aging	3.3.2	4.4.2	S-1	1.0

4.3 Examination. The filler shall be examined to determine compliance with requirements contained in this specification.

4.3.1 Dimensional. The filler shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.2 For defects. Examination shall be conducted in accordance with table III .

TABLE III. Classification of defects

Classification	Defect
Major A	
101	Filler not free of tears or holes.
102	Filler not free of deformation.
103	Filler not free of cracks.
104	Filler not free of tackiness.
105	Filler not resilient.
106	Bonded parts not free of separation (while being squeezed).
107	Filler material not as specified.
Major B	
151	Bulb cannot be evacuated.
152	Vacuum cannot be released through filling or discharge valves, with the pipet receptacle plugged.
Minor	
201	Markings not present, complete, correct, legible, or permanent.
202	Instructions not present, complete, correct, or legible.

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4.3.3 Examination of preparation for delivery. The packaging, packing, marking, and unitization (when supplied) shall be examined to determine compliance with the requirements of section 5. The sample unit shall be one container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Ball valves. Each ball valve shall be tested individually at a pressure of 20 pounds per square inch gage (psig). Leakage shall be determined by the use of an air flow meter. The flow meter shall have graduations of 0.01 cubic foot per hour and a maximum capacity not exceeding 0.10 standard cubic feet per hour.

4.4.2 Aging. The rubber of the pipet filler shall be tested as specified in method 7111 of Fed. Test Method Std. No. 601. NOTE: The determination of tensile strength and elongation in the aged specimen is not required.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. The pipet filler shall be wrapped in tissue paper, or enclosed in a plastic bag of appropriate size and further packaged in a paperboard box. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Twelve unit packages shall be packaged in a paperboard box. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.2 Commercial. The pipets shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, or commercial as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 fillers) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. The box shall be closed, waterproofed and banded in accordance with the appendix of PPP-B-636.

5.2.2 Commercial. The pipets shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking. Intermediate packages, shipping containers, and unitized loads (when applicable) shall be marked as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The pipet filler covered by this specification is intended for use in transferring corrosive, toxic, infectious and odorous liquids when used with a suitable pipet.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) National stock number (NSN).
(c) Color required, if other than black (see 3.4).
(d) Packaging and packing (see 5.1 and 5.2).
(e) Marking required (see 5.3).

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6.3 Flowmeter. A flowmeter suitable for conducting the ball valve leakage test specified in 4.4.1 is SGA Scientific Inc., Cat. No. G-6382 (Matheson Model No. 7261).

6.4 This specification is intended to cover the following item:

National stock number	Item identification
6640-00-889-1712	FILLER, PIPET, RUBBER (BLACK)
6640-00-045-1608	FILLER, PIPET, RUBBER (RED) For use with enzyme solutions

MILITARY CUSTODIANS:

Army - MD
Navy - MS
Air Force - 03

Military Coordinating Activity:

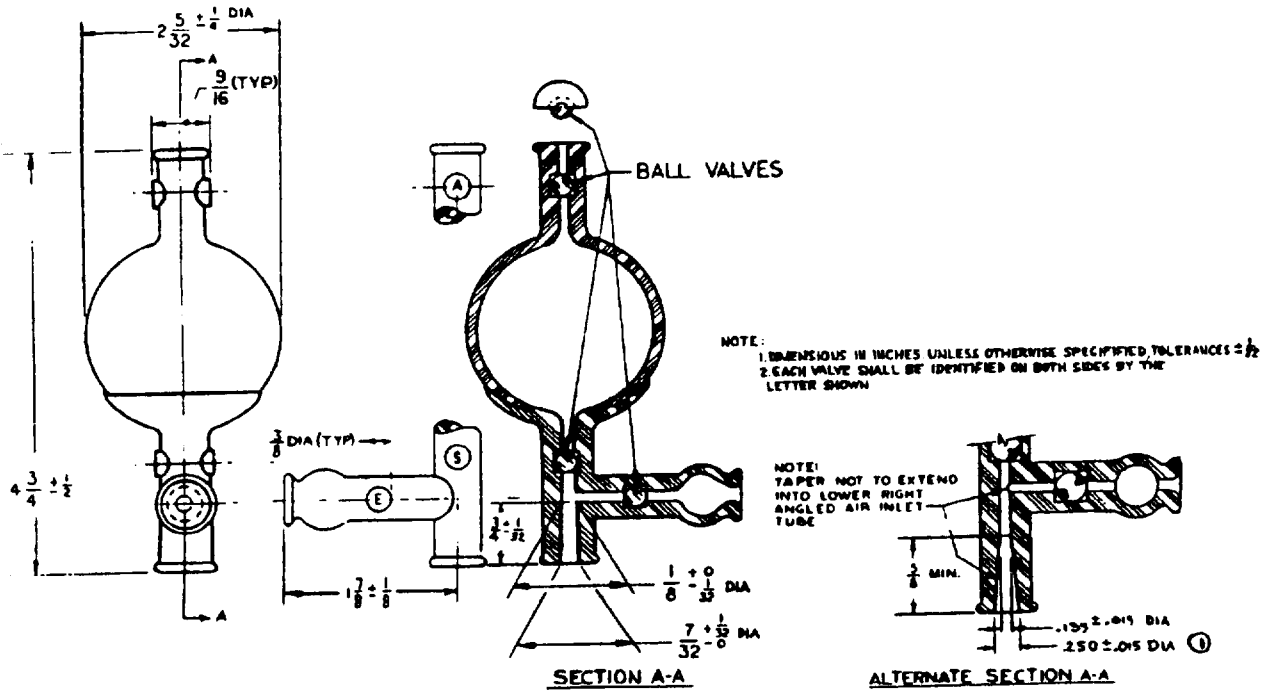
DSA-DM

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-DMS



(1) At the theoretical intersection of the conical internal surface and the plane of the bottom vertical tube.

FIGURE 1. Filler, pipet, rubber

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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